Grade 9 Physics

Illustrated study notes · Electricity · Forces · Magnetism · Electrostatics
November 2026 exam (1½ hours, 75 marks)

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1. Formula sheet and how to do calculations
2. Electricity: cells, energy, generating electricity, the national grid, cost
3. Electricity: circuit components, current, voltage, resistance
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6. Forces: types of forces, force diagrams, mass and weight
7. Magnetism: poles, magnetic fields, magnetic materials, Earth's field
8. Electrostatics: charges, charging by rubbing, conservation of charge

Also learn the definitions on your class definition page. These notes use **g = 10 N/kg**; use 9.8 N/kg if your teacher does.

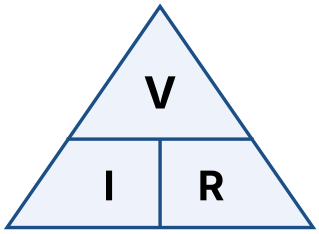
1. Formula sheet

Every calculation in this paper uses one of these formulas. Learn the **symbol** and the **unit** of each quantity.

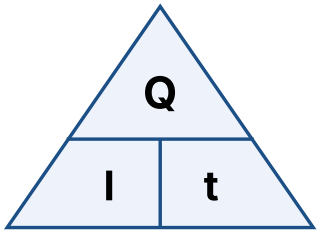
Quantity	Symbol	Unit	Formula	Rearranged
Charge	Q	coulomb (C)	**Q = I × t**	I = Q ÷ t t = Q ÷ I
Potential difference (voltage)	V	volt (V)	**V = I × R**	I = V ÷ R R = V ÷ I
Energy delivered by a cell	W	joule (J)	**W = V × Q**	V = W ÷ Q Q = W ÷ V
Weight	w	newton (N)	**w = m × g**	m = w ÷ g
Electrical energy used	E	kilowatt-hour (kWh)	**E = P (kW) × t (h)**	
Cost of electricity	–	rand (R)	**cost = kWh × price per kWh**	

The triangle trick

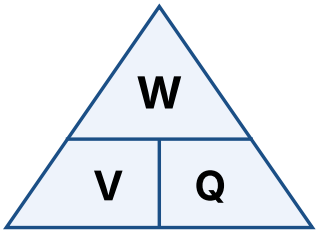
All these formulas have the form **A = B × C**. Put A on top of a triangle and B and C at the bottom. Cover the one you want: if the other two are **side by side**, multiply them; if one is **above** the other, divide.



V = I × R · I = V ÷ R · R = V ÷ I



Q = I × t · I = Q ÷ t · t = Q ÷ I



W = V × Q · V = W ÷ Q · Q = W ÷ V

Unit conversions you must do first

Change	How	Example
minutes → seconds	× 60	3 min = 180 s
hours → seconds	× 3 600	2 h = 7 200 s
milliampere (mA) → ampere (A)	÷ 1 000	250 mA = 0.25 A
gram (g) → kilogram (kg)	÷ 1 000	500 g = 0.5 kg
watt (W) → kilowatt (kW)	÷ 1 000	2 000 W = 2 kW

How to get full marks for a calculation: (1) write the **formula**, (2) **substitute** the numbers in the correct units, (3) write the **answer with its unit**. A missing unit loses a mark.

Example: A current of 0.5 A flows for 2 minutes. Calculate the charge.

t = 2 × 60 = 120 s

Q = I × t

Q = 0.5 × 120

Q = 60 C

2. Cells, energy and generating electricity

What is a cell?

A **cell** is a small store of **chemical potential energy**. When it is connected in a **closed circuit**, chemicals inside it react and push charges (electrons) around the circuit. In this way the cell changes **chemical energy → electrical energy**.

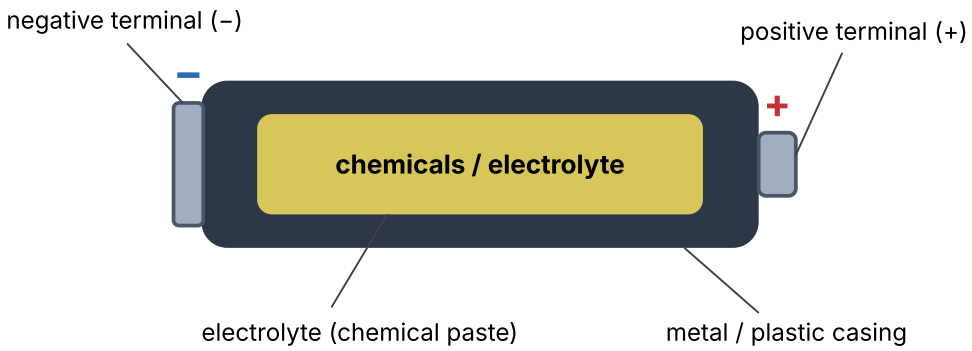


Figure 1: Inside a dry cell (like an AA torch battery, 1.5 V)

- The chemicals only react when the circuit is **closed** – that is why a cell lasts longer when the switch is off.
- When all the chemicals have reacted, the cell is **flat** and can't push charge any more.
- The **voltage** of a cell tells you how much energy it gives every coulomb of charge: a 1.5 V cell gives **1.5 J to every 1 C**.

Cells vs batteries

A **battery** is **two or more cells connected together**. When cells are joined in series (+ to −), their voltages **add up**.

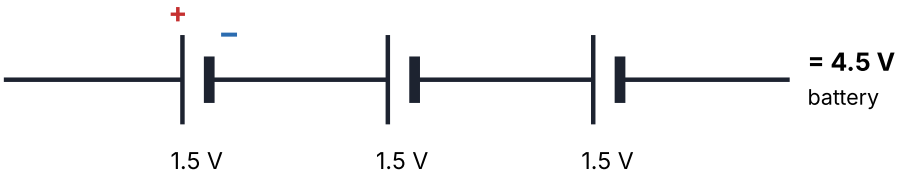


Figure 2: Three 1.5 V cells in series make a 4.5 V battery. Long thin line = positive (+); short thick line = negative (−).

Energy changes in a circuit

Energy is **never created or destroyed** – it only changes from one form to another. In a torch: chemical (cell) → electrical (wires) → light + heat (bulb).

Device	Energy change
Cell / battery	chemical → electrical
Light bulb	electrical → light + heat
Kettle, heater, iron, toaster	electrical → heat
Fan, electric motor, drill	electrical → kinetic (movement)
Buzzer, speaker, radio	electrical → sound
Charging a phone	electrical → chemical (stored in the battery)
Solar panel	light → electrical

Energy delivered by a cell: W = V × Q
A 1.5 V cell gives a current of 0.2 A for 5 minutes. How much energy does it deliver?
Step 1 – charge: Q = I × t = 0.2 × 300 = 60 C
Step 2 – energy: W = V × Q = 1.5 × 60 = **90 J**

How a power station makes electricity

Almost all power stations work the same way: something turns a **turbine**, and the turbine turns a **generator**. Inside the generator, **magnets spin near coils of copper wire**, and this movement produces an electric current.

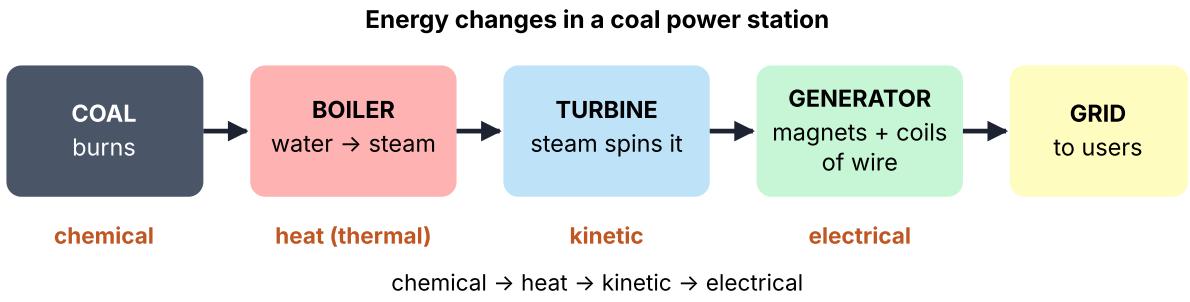


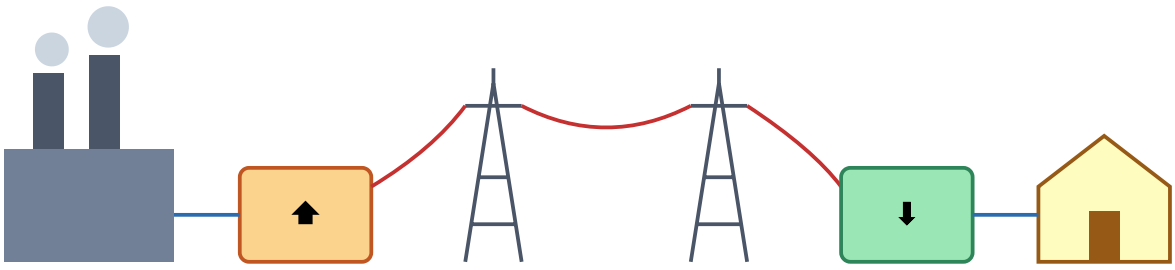
Figure 3: A coal-fired power station. Most of South Africa's electricity is made this way (Eskom).

Methods of generating electricity

Method	What turns the turbine?	Energy change	Renewable?	Good / bad
Coal	Steam from burning coal	chemical → heat → kinetic → electrical	No	+ SA has lots of coal, works day and night – air pollution, CO_2 (global warming), uses lots of water
Nuclear (Koeberg, Cape Town)	Steam heated by splitting uranium atoms	nuclear → heat → kinetic → electrical	No	+ no CO_2, lots of energy – dangerous radioactive waste, expensive
Hydro-electric	Falling water from a dam	gravitational potential → kinetic → electrical	Yes	+ clean – needs big rivers; SA is dry
Wind	Wind turns the blades	kinetic → electrical	Yes	+ clean – only when the wind blows
Solar	No turbine – solar panels change light directly	light → electrical	Yes	+ clean, lots of sun in SA – only in daylight, panels are costly

The national grid

The **national grid** is the huge network of power lines, transformers and substations that carries electricity from power stations to homes, schools and factories all over South Africa.



High voltage → small current → less energy lost as heat in the long wires

Figure 4: How electricity gets from the power station to your home

Why such a high voltage? At a high voltage, the same energy can be sent with a much **smaller current**. A smaller current heats the wires less, so **less energy is wasted** over hundreds of kilometres. The voltage is then lowered (stepped down) to a safe 230 V before it reaches homes.

Load shedding happens when people need more electricity than the power stations can make. Eskom switches areas off in turns so that the whole grid does not collapse.

Cost of electricity

- **Power** is how fast an appliance uses energy, in watts (W). 1 kilowatt (kW) = 1 000 W.
- Electricity is sold in **kilowatt-hours (kWh)**, also called **units**. 1 kWh = using 1 kW for 1 hour.
- **Energy (kWh) = power (kW) × time (hours)** and **Cost = kWh × price per kWh**

Example: A 2 000 W heater is on for 3 hours. Electricity costs R2.50 per kWh.
Power = 2 000 ÷ 1 000 = 2 kW
Energy = 2 kW × 3 h = 6 kWh
Cost = 6 × R2.50 = **R15.00**

Saving electricity: appliances that make **heat** (geyser, kettle, heater, stove) use the most. Switch off lights and appliances, use LED bulbs, put a blanket on the geyser, use a solar water heater, and boil only the water you need.

3. Circuits: components and key terms

An **electric circuit** is a **closed path** that charge can flow around. It needs an energy source (cell), connecting wires, and something that uses the energy (bulb, resistor). If there is a gap anywhere – an open switch or a broken wire – it is an **open circuit** and no current flows.

Circuit symbols

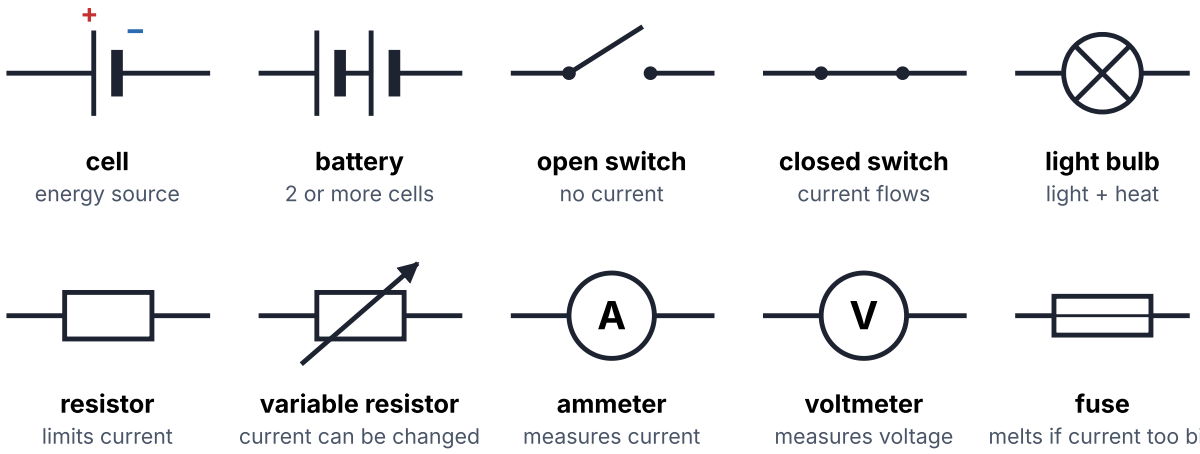


Figure 5: Circuit symbols you must know. Draw circuits with a ruler, straight lines and square corners.

A simple circuit

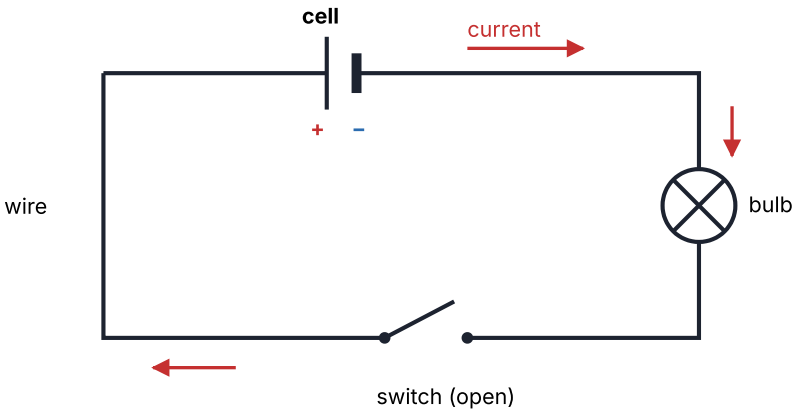


Figure 6: A cell, a switch and a bulb. The switch is open, so no current flows yet. When it closes, conventional current (red arrows) flows from + to – around the outside of the circuit.

The four key terms

Term	What it means (easy words)	Unit	Measured with
Charge (Q)	The "stuff" that flows in a wire – carried by electrons. Charge is **not used up**; it goes round and round. It is the **energy** that gets used.	coulomb (C)	–
Current (I)	How much charge flows past a point **every second** (rate of flow of charge).	ampere (A)	**Ammeter**, connected **in series**
Potential difference (V) (voltage)	The **energy given to each coulomb** of charge between two points. Think of it as the "push" of the cell.	volt (V)	**Voltmeter**, connected **in parallel** (across a component)
Resistance (R)	How much a component **opposes (slows down) the current**. More resistance = less current.	ohm (Ω)	Calculate: R = V ÷ I

Water pipe picture: the cell is like a pump, the **voltage** is how hard the pump pushes, the **current** is how much water flows per second, and **resistance** is a narrow part of the pipe that makes it harder for water to flow.

Example (Q = I × t): 120 C passes a point in 1 minute. Find the current.
t = 60 s → I = Q ÷ t = 120 ÷ 60 = **2 A**

Example (V = I × R): A voltmeter across a bulb reads 9 V and the ammeter reads 1.5 A. Find the resistance.
R = V ÷ I = 9 ÷ 1.5 = **6 Ω**

Example (V = I × R): A 12 V battery is connected to a 4 Ω resistor. Find the current.
I = V ÷ R = 12 ÷ 4 = **3 A**

4. Series and parallel circuits

In a **series** circuit there is only **one path** for the current. In a **parallel** circuit there are **two or more paths (branches)**.

Series circuit

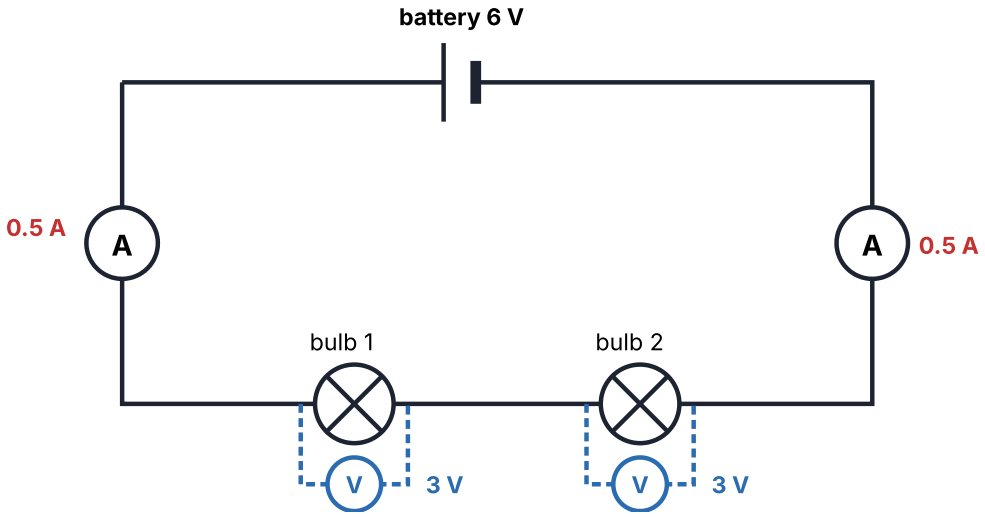


Figure 7: A series circuit. Both ammeters read the same (0.5 A). The 6 V is shared between the bulbs: 3 V + 3 V = 6 V.

Rule for series circuits	In symbols
The **current is the same everywhere** (only one path, so all the charge goes through everything)	$I = I_1 = I_2$
The **voltage is shared** between the components; the parts add up to the battery's voltage	$V = V_1 + V_2$
The **total resistance adds up**	$R = R_1 + R_2$
Add more bulbs → more resistance → less current → bulbs get **dimmer**	
One bulb breaks → the circuit is open → **all** the bulbs go out	

Series example: A 12 V battery with a 2 Ω and a 4 Ω resistor in series.
(a) Total resistance: R = 2 + 4 = **6 Ω**
(b) Current: I = V ÷ R = 12 ÷ 6 = **2 A** (same through both resistors)
(c) Voltage across 2 Ω: V = I × R = 2 × 2 = **4 V**
(d) Voltage across 4 Ω: V = 2 × 4 = **8 V** Check: 4 + 8 = 12 V ✓

Parallel circuit

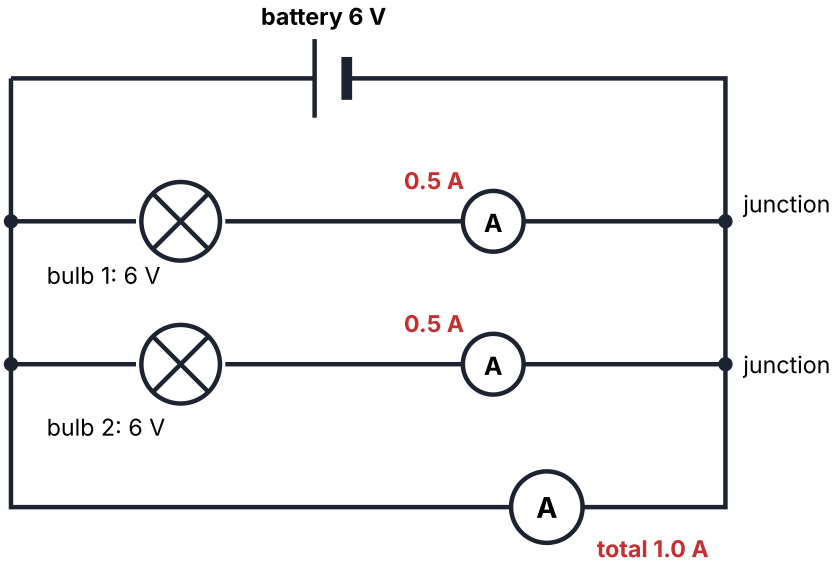


Figure 8: A parallel circuit with two branches. Each bulb gets the full 6 V. The current splits: 0.5 A + 0.5 A = 1.0 A from the battery.

Rule for parallel circuits	In symbols
The **voltage is the same** across every branch (each branch is connected straight to the battery)	$V = V_1 = V_2$
The **current splits** at a junction; the branch currents add up to the total	$I = I_1 + I_2$
More current flows through the branch with the **smaller resistance**	
Adding more resistors in parallel makes the **total resistance smaller** (more paths for the current)	$1/R = 1/R_1 + 1/R_2$
Add more bulbs → each stays **equally bright** (but the battery goes flat faster)	
One bulb breaks → the **others stay on**. That is why houses are wired in parallel.	

Parallel example: A 6 V battery with a 3 Ω and a 6 Ω resistor in parallel.
(a) Voltage across each resistor = **6 V**
(b) Current through 3 Ω: $I = V \div R = 6 \div 3 =$ **2 A**
(c) Current through 6 Ω: $I = 6 \div 6 =$ **1 A**
(d) Total current: 2 + 1 = **3 A**
(e) Total resistance: $R = V \div I = 6 \div 3 =$ **2 Ω** (smaller than 3 Ω ✓)

Example: The main ammeter reads 1.2 A and the ammeter in branch 1 reads 0.5 A. Find the branch 2 current.
$I = I_1 + I_2 \rightarrow 1.2 = 0.5 + I_2 \rightarrow I_2 =$ **0.7 A**

Series vs parallel at a glance

	Series	Parallel
Paths	One	More than one
Current	Same everywhere	Splits between branches
Voltage	Shared (adds up)	Same across each branch
Total resistance	Gets bigger as you add resistors	Gets smaller as you add resistors
One bulb blows	All go out	Others stay on
Example	Old Christmas lights	House wiring

5. What affects the resistance of a wire?

Resistance happens because moving electrons **bump into the atoms** of the wire. Anything that causes **more bumps** gives **more resistance**.

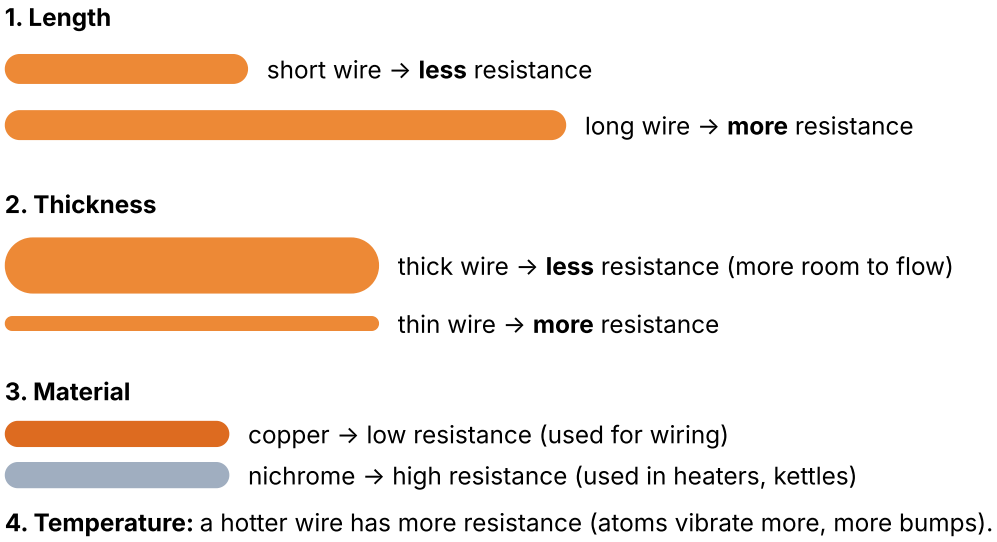


Figure 9: The four factors that affect resistance

- **Conductors** let current flow easily: metals (copper, aluminium, silver, iron), graphite, salt water.
- **Insulators** do not let current flow: plastic, rubber, wood, glass, dry air.
- Electric wires are **copper** (good conductor) covered in **plastic** (insulator, for safety).

> **Practical investigation:** to test how length affects resistance, change only the **length** (independent variable), measure the **current** (dependent variable), and keep the thickness, material, temperature and battery the **same** (controlled variables). Result: longer wire → lower ammeter reading → higher resistance.

6. Forces

A **force** is a **push or a pull** on an object. A force always has a **size** and a **direction**, so we draw it as an **arrow**. Force is measured in **newtons (N)** with a spring balance. (1 N is about the weight of a small apple.)

The 4 effects of forces

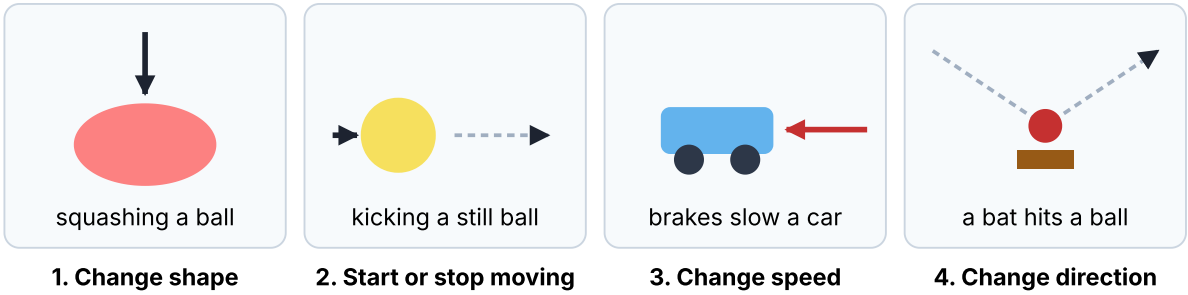
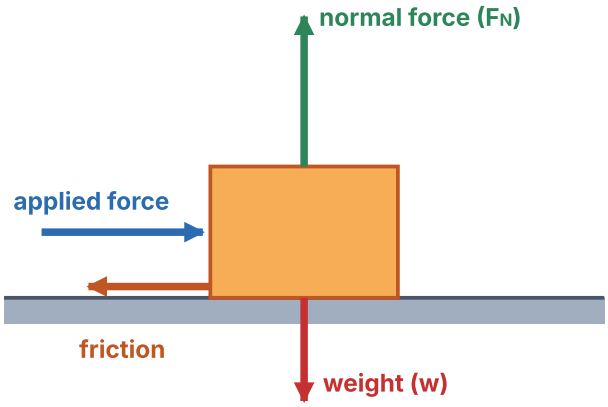


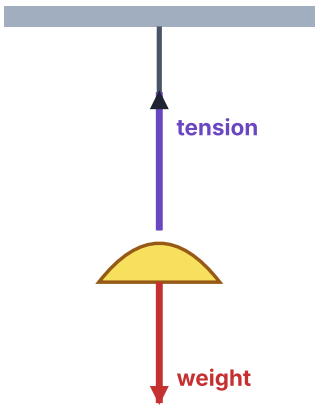
Figure 10: A force can change the shape, motion, speed or direction of an object

Contact forces and field forces

Contact forces only act when objects **touch**. **Field forces** (non-contact forces) act **from a distance**, without touching.



A box pushed to the right: friction acts to the left, against the motion



A hanging lamp: tension pulls up along the cable

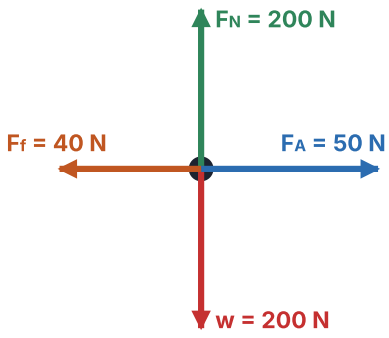
Contact force	Explanation	Direction
Applied force	A push or pull by a person or object (pushing a trolley)	The way you push or pull
Friction	Force between two surfaces rubbing together that **opposes motion**. Rough surfaces → more friction. It produces heat.	**Opposite** to the motion, along the surface
Normal force	The push of a surface on an object resting on it (a table pushing up on a book)	**Perpendicular (90°)** to the surface
Tension	The pull in a stretched rope, string or cable	Along the rope, away from the object

Field force	Explanation
Weight (gravitational force)	The Earth pulls every object towards its centre – always drawn **straight down**
Electrostatic force	Attraction or repulsion between **charged** objects (section 8)
Magnetic force	Attraction or repulsion between **magnets**, or a magnet and iron (section 7)

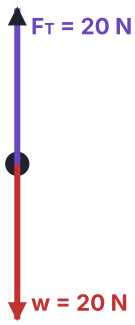
> **Friction is useful:** it lets us walk, gives tyres grip and makes brakes work. **Friction is a nuisance:** it wears out shoes and machine parts and wastes energy as heat. Reduce it with oil, grease, ball bearings and smooth or streamlined shapes.

Free body diagrams

A **free body diagram** shows the object as a **dot**, with an arrow for every force acting **on** it, drawn **from the dot**.



A box pushed right and speeding up: the applied force arrow is longer than friction



The hanging lamp: equal arrows, because the forces balance

Rules for force diagrams: (1) only forces acting **on** the object, (2) arrows point in the direction of the force, (3) **longer arrow = bigger force**, equal forces get equal arrows, (4) label every arrow (w, F_N, F_A, F_f, F_T), (5) use a ruler.

Mass and weight

Mass is the **amount of matter** in an object, measured in **kilograms (kg)**. **Weight** is the **force with which the Earth pulls** on the object, measured in **newtons (N)**. Your mass is the same everywhere, but your weight depends on how strong gravity is.

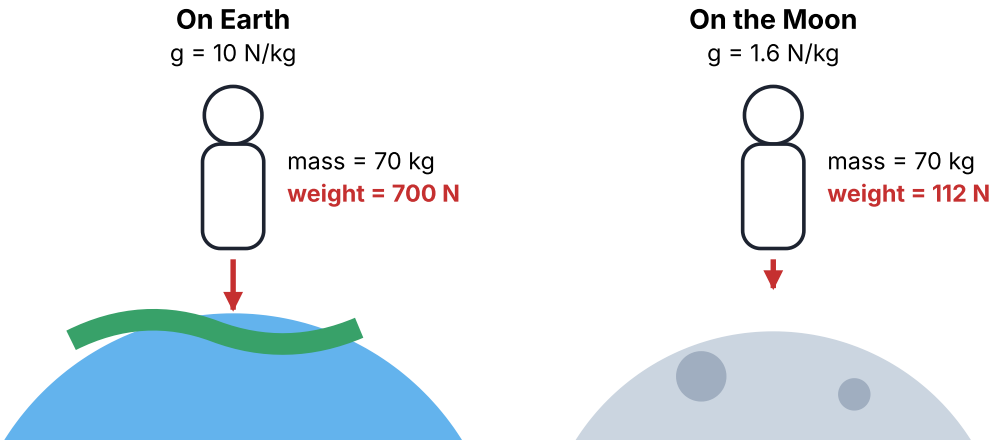


Figure 11: An astronaut has the same mass everywhere, but weighs about ⅙ as much on the Moon because the Moon's gravity is weaker

	Mass	Weight
Meaning	Amount of matter	Gravitational pull of the Earth on an object
Unit	kilogram (kg)	newton (N)
Measured with	Balance / scale	Spring balance
Is it a force?	No	Yes – a field force, acting downwards
Changes on the Moon?	No	Yes – about ⅙

w = m × g (g = 10 N/kg on Earth)
Weight of a 45 kg learner: w = 45 × 10 = **450 N**
Weight of a 250 g packet: m = 0.25 kg → w = 0.25 × 10 = **2.5 N**
Mass of a box that weighs 800 N: m = w ÷ g = 800 ÷ 10 = **80 kg**

In everyday life people say "I weigh 50 kg". In science, 50 kg is your **mass**; your weight is 500 N.

7. Magnetism

A **magnet** attracts magnetic materials such as iron. The **magnetic force** is a **field force** – it works without touching, and it gets weaker the further away you go.

Poles of a magnet

- Every magnet has a **north pole (N)** and a **south pole (S)**. The force is **strongest at the poles**.
- **Like poles repel** (N–N, S–S) and **unlike poles attract** (N–S).
- Poles always come in pairs. Cut a magnet in half and you get **two** smaller magnets, each with an N and an S pole.
- A hanging magnet (or a compass needle) turns until its N pole points **north**.
- **Repulsion is the only sure test for a magnet** – a magnet attracts iron *and* other magnets, but it can only repel another magnet.

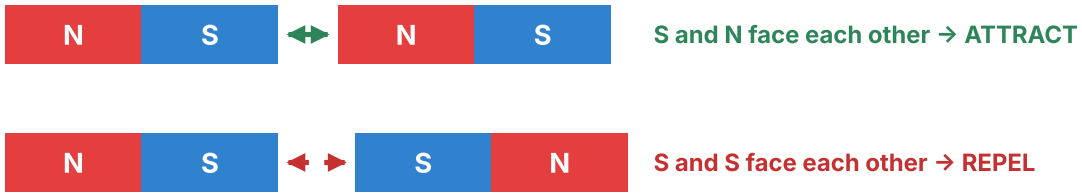


Figure 12: Unlike poles attract, like poles repel

Magnetic fields

A **magnetic field** is the area around a magnet where a magnetic force can be felt. You can see its shape by sprinkling **iron filings** around a magnet, or by moving a small **plotting compass** around it.

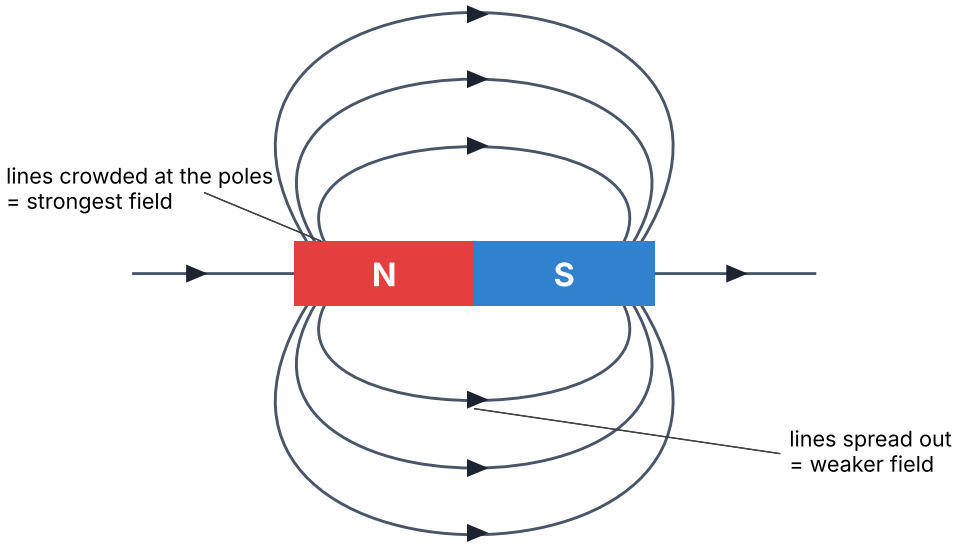
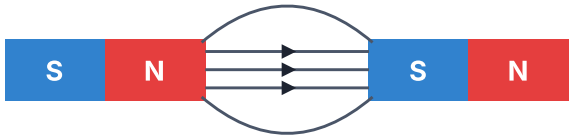
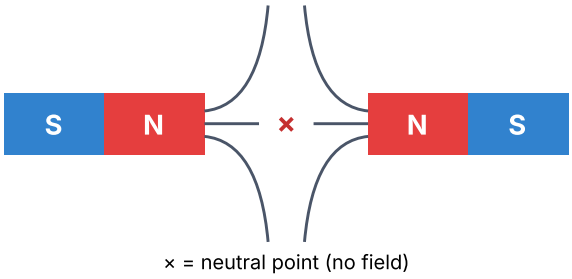


Figure 13: Magnetic field of a bar magnet. The arrows go from N to S outside the magnet.

Rules for drawing field lines: (1) arrows point **from N to S** outside the magnet, (2) lines **never cross**, (3) lines are **closest together at the poles**, where the field is strongest.



Unlike poles (N facing S): lines go straight across – attraction

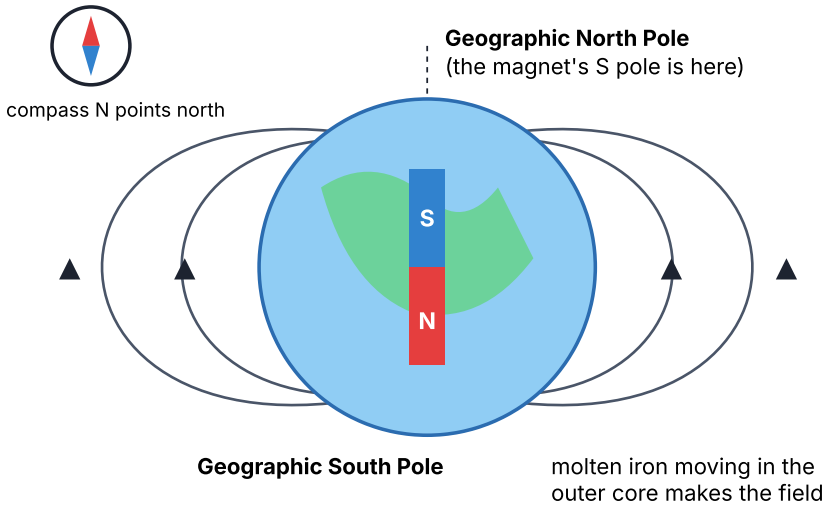


Like poles (N facing N): lines bend away – repulsion

Magnetic and ferromagnetic materials

- **Magnetic materials** are attracted to a magnet. **Ferromagnetic materials** are **strongly** attracted and can be made into magnets: **iron, nickel, cobalt** and their alloys such as **steel**.
- **Non-magnetic materials:** wood, plastic, glass, paper, rubber, and most metals – **copper, aluminium, gold, silver, brass**.
- ⚠ **Not all metals are magnetic.** An aluminium cold-drink can is not attracted; a steel food can is.
- **Soft iron** magnetises easily but loses its magnetism quickly (temporary magnet – used in electromagnets). **Steel** keeps its magnetism (permanent magnet – fridge magnets, compass needles).
- **To make a magnet:** stroke a steel nail many times in **one direction** with one pole of a magnet. **To destroy it:** heat it, hammer it or drop it.

The Earth's magnetic field



Field lines point up: out of the magnet's N pole (south) and into its S pole (north)

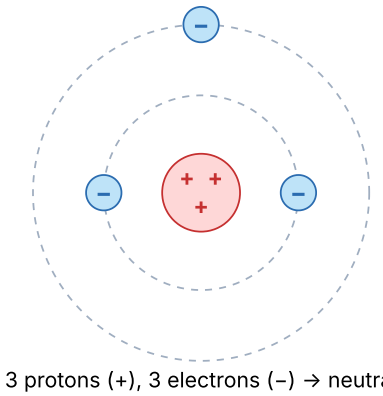
Figure 14: The Earth acts like a giant bar magnet. Its magnetic south pole lies near the geographic North Pole – so the N end of a compass is pulled north (unlike poles attract).

- The field is caused by **moving molten iron and nickel in the Earth's outer core**.
- The magnetic poles are not exactly at the geographic poles, and they slowly move.
- The magnetic field (the **magnetosphere**) **protects us from the solar wind** – harmful charged particles from the Sun.
- Near the poles some particles get in and make the sky glow: the **aurora** (northern and southern lights).
- Uses: navigating with a **compass**; birds and turtles use the field to find their way when they migrate.

8. Electrostatics

Electrostatics is about **static electricity** – charges that build up on objects and stay there. Objects become charged when **electrons move** from one object to another.

Charges in an atom



A neutral atom has equal + and – charges

Particle	Charge	Where	Can it move to another object?
Proton	+ positive	Nucleus (centre)	**No** – stuck in the nucleus
Neutron	none	Nucleus	No
Electron	– negative	Around the nucleus	**Yes** – the outer ones can be rubbed off

A **neutral** object has the same number of protons and electrons. Charge is measured in **coulomb (C)**.

How objects become charged: rubbing (friction)

When two different materials are **rubbed together**, **electrons move from one to the other**.

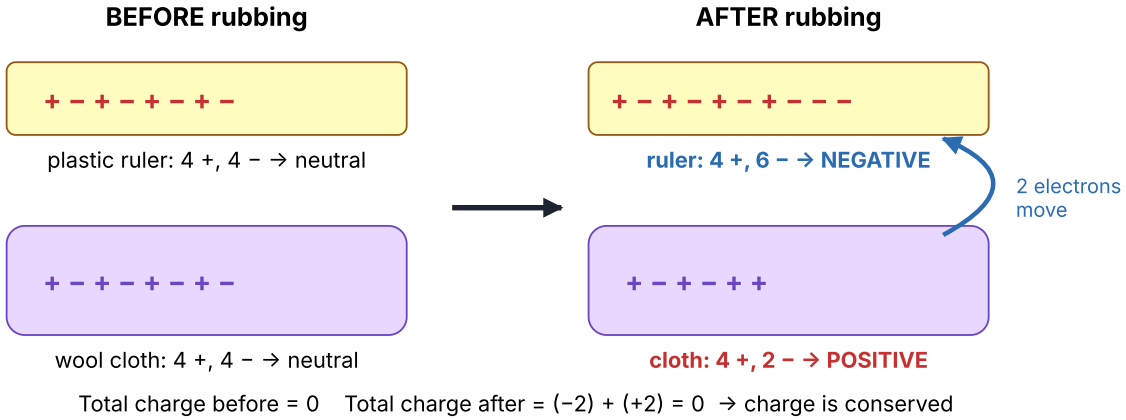


Figure 15: Rubbing a plastic ruler with a woollen cloth. Electrons move from the cloth to the ruler.

The object that...	ends up with...	so its charge is
gains electrons	more electrons than protons	**negative (–)**
loses electrons	fewer electrons than protons	**positive (+)**

Only electrons move. An object becomes positive by **losing electrons** – never by gaining protons. The two objects always end up with **opposite charges of the same size**.

Rubbed together	Becomes negative (gains e^-)	Becomes positive (loses e^-)
Plastic ruler + wool	Plastic ruler	Wool
Glass rod + silk	Silk	Glass rod
Balloon + hair	Balloon	Hair

The electrostatic force

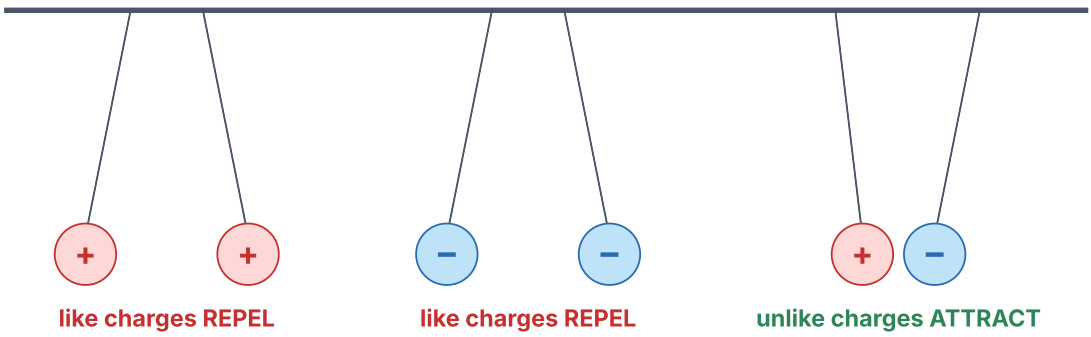


Figure 16: Charged balls hanging on threads

- The **electrostatic force** is the attraction or repulsion between charged objects. It is a **field force**.
- **Like charges repel**; **unlike charges attract**.
- The force is **stronger** when the objects are **closer** and when they carry **more charge**.
- A charged object also **attracts neutral objects**: a charged comb picks up small bits of paper, and a rubbed balloon sticks to a wall.

Conservation of charge

> **Charge cannot be created or destroyed – it can only be moved (transferred) from one object to another.** The total charge before and after rubbing is the same. In Figure 15, the ruler gained 2 electrons and the cloth lost exactly 2 electrons, so the total stays 0.

Static electricity in everyday life

- A small shock from a car door or after walking on a carpet; hair standing up after taking off a jersey.
- **Lightning** – charge builds up in thunderclouds and jumps to the ground as a giant spark.
- **Uses:** photocopiers and printers, spray-painting cars, dust filters in factory chimneys.
- **Dangers:** sparks can light petrol or gas – fuel trucks are **earthed** with a chain or cable. **Lightning conductors** on tall buildings carry charge safely into the ground.